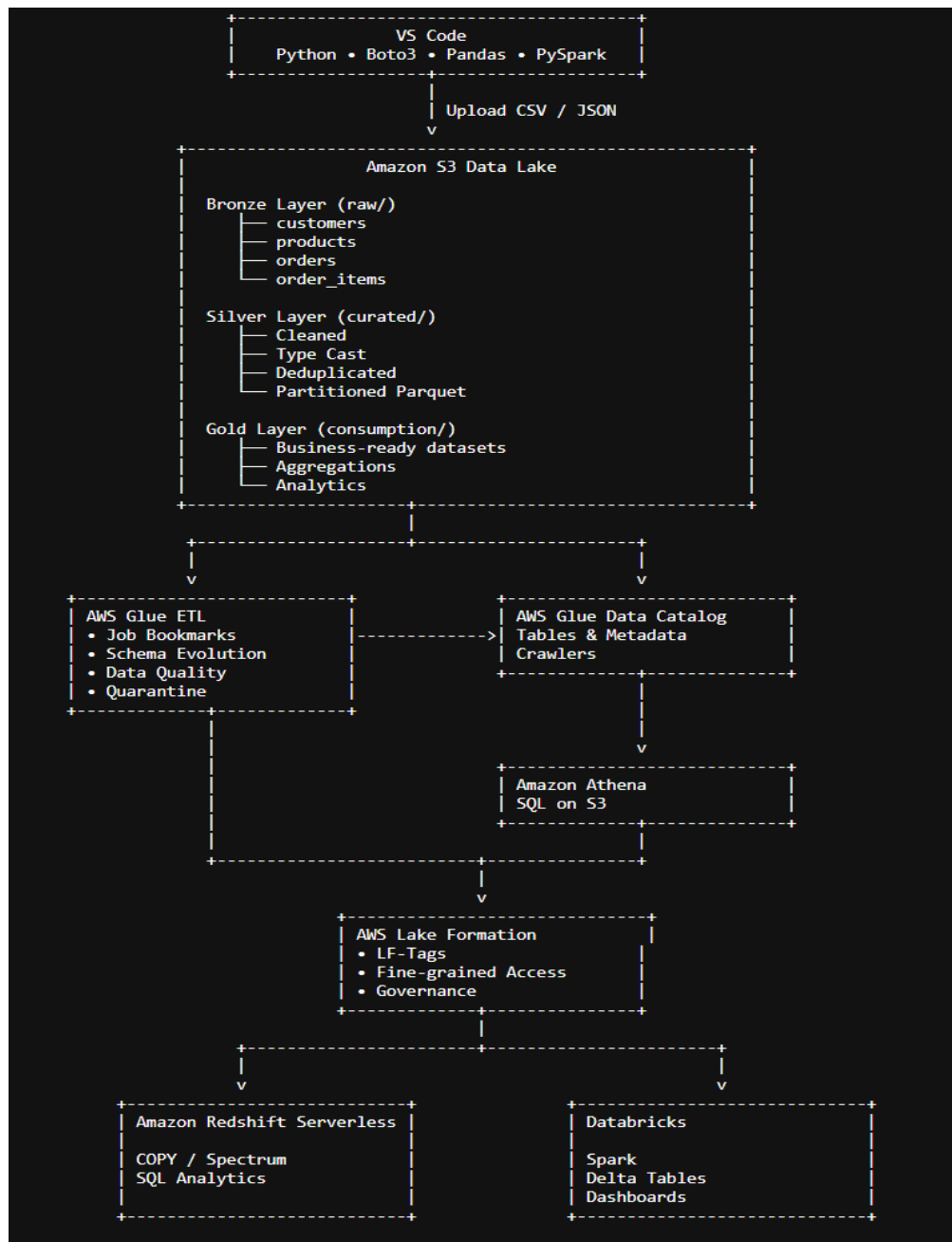


1) Data Engineering Lakehouse Architecture



2) AWS Pipeline Cost & Infrastructure Review

AWS Region: us-east-1 (N. Virginia)

Dataset Scale: 15.62 MB (Baseline) → 156.24 MB (10x Volume)

1. Provisioned Architecture (Active Build)

During the development phase, four primary AWS services were deployed to process the multi-source dataset (clickstream logs, customers, orders, and products):

- **Amazon S3:** Raw ingestion (raw-zone) and Parquet storage (analytics-zone).
- **AWS Glue:** Schema discovery via Crawlers and data transformation via 2-DPU PySpark jobs.
- **Amazon Athena:** Serverless SQL engine used for schema validation and ad-hoc data checks.
- **Amazon Redshift:** Provisioned single-node dc2.large cluster for warehousing and BI queries.

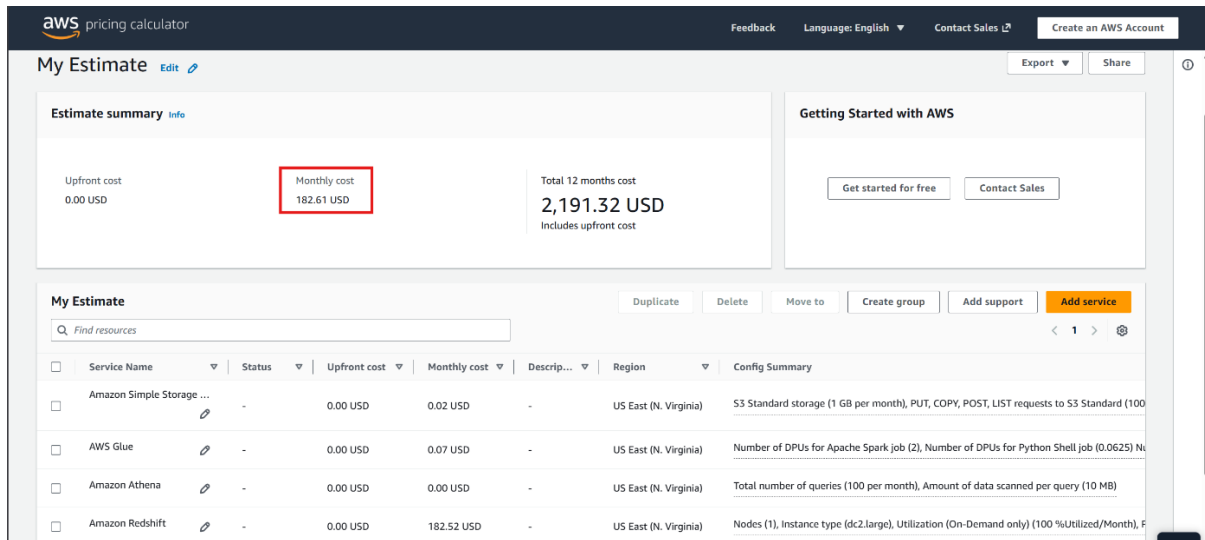
2. Decommissioning & Teardown Verification

To enforce cloud cost management and prevent idle resource charges after development, all resources were torn down:

- **Amazon Redshift:** Terminated dc2.large cluster (snapshot created prior to deletion).
- **AWS Glue:** Deleted Spark jobs, triggers, and Glue Catalog schemas.
- **Amazon S3:** Cleared all raw/processed objects and deleted S3 buckets.
- **Amazon Athena:** Saved query history cleared and result location emptied.

3. Monthly Cost Review at 10x Scale (156.24 MB)

Below is the monthly cost breakdown for running the pipeline at 10x volume, calculated using the **AWS Pricing Calculator**:



4. Cost Optimization Recommendations

- **Key Driver:** Redshift cluster hosting accounts for **99.95% (\$182.52)** of the total monthly expenditure.
- **Switch to Redshift Serverless:** Since batch loads run intermittently, using Redshift Serverless will reduce Redshift costs from **\$182.52** to **~\$0.50/month**.
- **Use S3 + Athena (Data Lake):** For small to medium datasets (<100 GB), storing data as Parquet in S3 and querying serverless via Athena drops the entire pipeline cost to **under \$0.10/month**.